

Jacob C Blevins

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

- PhD – Mechanical Engineering: Robotics & Control May 2022 – Expected: Dec 2026
 - Advisor: Dr. Jun Ueda, Biorobotics and Human Modeling Lab
- MS – Computer Science: Machine Learning May 2024 – Expected: Dec 2025
 - Advisor: Dr. Sehoon Ha, Ha Lab
- MS – Mechanical Engineering: Design Aug 2020 – May 2021
- BS – Mechanical Engineering Aug 2016 – May 2020

RESEARCH & PUBLICATIONS

In Progress

- Reward as Observation: Learning Reward-based Policies for Rapid Adaptation
- Affine Perfectly Undetectable False Data Injection Attacks on Multi-robot Formation Control
- Deep learning for affine transformation-based perfectly undetectable false data injection attacks to general systems

Completed

- H. Kwon, J. Blevins and J. Ueda, "Defense Mechanisms Against Undetectable Cyberattacks on Encrypted Telerobotic Control Systems," in *IEEE/ASME Transactions on Mechatronics*, vol. 30, no. 4, pp. 2964-2971, Aug. 2025.
- J. Blevins and J. Ueda, "Encrypted Model Reference Adaptive Control With False Data Injection Attack Resilience via Somewhat Homomorphic Encryption-Based Overflow Trap," in *IEEE Transactions on Industrial Cyber-Physical Systems*, vol. 3, pp. 262-272, 2025
- Jacob Blevins, Amit Jariwala. "Leveraging AI Chatbots in Makerspaces: Enhancing Learning and Collaboration." *ISAM*, 2024.
- J. Ueda and J. Blevins, "Affine Transformation-Based Perfectly Undetectable False Data Injection Attacks on Remote Manipulator Kinematic Control With Attack Detector," in *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8690-8697, Oct. 2024.
- H. B. Kwon, S. Kosieradzki, J. Blevins and J. Ueda, "Encrypted Coordinate Transformation via Parallelized Somewhat Homomorphic Encryption for Robotic Teleoperation," 2023 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), Seattle, WA, USA, 2023, pp. 228-233.

PROJECTS

- *Vision Language Frontier Maps Plus (VLFM+)* Present
 - Improving the VLFM (Boston Dynamics) pipeline with improved models and diffusion-based trajectory optimization
- *Domain Randomization for Robot Locomotion* Jan 2025 – May 2025
 - Utilizing deep reinforcement learning and domain randomization for robust quadruped locomotion in novel environments
- *Reinforcement Learning Toolbox* Present
 - Custom toolbox for reinforcement learning training and inference on single or multiagent systems
- *NVIDIA Open Hackathon: Language to Action* Jan 2025 – May 2025
 - Llama3.2 paired with YOLOworld to auto generate and execute code for pathing and control commands from human language. i.e. tell your robot what to do and it will do it!
- *Robust Adversarial Learning for Multiagent Systems* Jan 2025 – May 2025
 - Multiagent deep reinforcement adversarial learning for coordinated motion planning under attacked environments.
- *Building a Vision-Based Object Tracking Model for Autonomous Vehicles* Sept 2024 – Dec 2024
 - Development of a vision-based deep perception model for object detection and tracking in road-way scenarios for autonomous vehicles.
 - A study on how variations in model architecture such as the addition of attention mechanisms affect final inference for this task.
- *Autonomous Mobile Robot via Machine Learning* May 2024 – July 2024
 - PPO, YOLOv8, K-means, and other ML algorithms collaborating to guide a Turtlebot3 through an unknown environment
- *RoboJackets - RoboNav Mars Rover Software Team* April 2024 – May 2025
 - Motion planning subteam – Development of path planning and control for a mars rover with an NVIDIA Jetson Orin Nano for traversal over complex terrain for the 2025 University Rover Competition
- *Liquid Sloshing Reduction via Input Shaping* Nov 2023
 - Input shaping of crane, double-pendulum, liquid system, reducing system modes to 5% of their original magnitude, saving factory workers from hazardous liquid sloshing

TECHNICAL SKILLS

- *Robotics* – ROS2, MuJoCo, Gazebo, RoboDK, IsaacSim, OpenAI Gym, OpenCV
 - *Programming* – Python, C/C++, MATLAB/Simulink
 - *Machine Learning* – Pytorch, Tensorflow, Scikit Learn, Tensorboard
 - *Design and Manufacturing* – CAD, 3D Printing, Laser cutting, Water Jetting, Metalworking, Woodworking, Welding
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EMPLOYMENT HISTORY

- *Georgia Institute of Technology – Associate Academic Professional* *Dec 2024 – Present*
 - Lecture System Dynamics, Experimental Methods, and Computing Techniques
 - Integrate ML techniques into the mechanical engineering curriculum, helping students understand how they can use data science to solve complex engineering problems
 - Manage undergraduate research and tutoring programs, bringing students to their full academic potential
- *Georgia Institute of Technology – Graduate Teaching Assistant and Researcher* *Aug 2020 – Present*
 - Research on security of networked robotic systems (see research & publications section)
 - Teaching and advising mechanical engineering laboratory focusing on heat transfer, thermodynamics, signals, systems ID, controls, IC engines, and refrigeration
- *Georgia Institute of Technology – Lab & Facilities Coordinator* *May 2021 – Nov 2024*
 - Teaching lab-based courses focusing on design, manufacture, and technical communication
 - Management and design of Georgia Tech’s mechanical engineering course labs, resulting in state-of-the-art machine and equipment availability and quality workflow for thousands of students and design teams
 - Training students on fabrication machinery, tools, and safety
- *AC & DC Power Technologies – Mechanical Engineer* *May 2020 – Aug 2020*
 - Design and Analysis of energy storage systems via AutoCAD drawings and MATLAB for validating failure modes
 - MATLAB programing of application for creating detailed customer power-load charts
- *Caterpillar (CAT) Large Power Systems Division – Large Engines Intern* *May 2019 – Aug 2019*
 - Large engine head fatigue testing, measurement, and temperature data analysis
- *Panasonic Automotive - Advanced Engineering Intern* *May 2018 – Aug 2018*
 - Redesigned and optimized the kinematics system of a General Motors Heads Up Display (GM HUD)
 - Lead the communications with suppliers as the key product engineer during the procurement of prototype parts
- *Georgia Institute of Technology – Fluid Mechanics Grader and Statics Tutor* *Jan 2018 – Dec 2019*

PRESENTATIONS

- *Ph.D. Proposal Presentation: False Data Injection Attacks to Robotics with Probabilistic Defense Mechanisms* *June 2025*
- *ICRA 2025: International Conference on Robotics and Automation: Affine Transformation-Based Perfectly Undetectable False Data Injection Attacks on Remote Manipulator Kinematic Control with Attack Detector* *May 2025*
- *Savannah River National Laboratory: Secure Control of Networked Dynamic Systems* *Sept 2024*

SERVICE & ORGANIZATIONS

Organizations

- FIRST Robotics Workshop Lead *Jan 2025*
- IEEE Robotics and Automation Society graduate student member *Aug 2024 – Present*
- RoboGrads at Georgia Tech *Aug 2024 – Present*

Professional Contributions

- Reviewer, *International Conference on Robotics and Automation* *Sept 2023, Sept 2024*
- Reviewer, *Modeling Estimation and Control Conference* *Mar 2024*

Georgia Institute of Technology Contributions

- Undergraduate Curriculum Committee – Mechanical Engineering *Jan 2025 – Present*
- Idea to Prototype Mentor *Jan 2025 – May 2025*
- Interactive Learning Committee – Mechanical Engineering *Aug 2022 – Jan 2025*

HONORS & AWARDS

- Tech to Teaching Certification – Georgia Tech *May 2025*
- GWW School of Mechanical Engineering Professional Support Excellence *May 2024*
- GWW School of Mechanical Engineering Culture Champion *May 2023*
- Highest Honors – Georgia Institute of Technology – GPA: 4.0 *Aug 2016 – May 2020*
- Eagle Scout (Boy Scouts of America) *June 2013*

EXTRACURRICULARS

- *Music* – Atlanta Symphony Orchestra bass vocalist and jazz vocalist
 - *Fitness* – Nationally competitive powerlifter, certified personal trainer, and CrossFit athlete
 - *Other* – PC and keyboard building, wood turning, juggling, unicycling, piano, and guitar
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